Dual Affine Spiral Orbits on $\mathbb{Z}^2$ Generated by Paired Unit Squares

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Abstract

We introduce a simple iterative geometric construction on the integer lattice $\mathbb{Z}^2$ consisting of paired unit squares that share a single corner. At each step, a new square is constructed outwardly on the hypotenuse of the isosceles right triangle formed by the outer edges adjacent to the shared corner. This process generates two interlocking affine spiral orbits on $\mathbb{Z}^2$: a positive orbit $\{P_n\}$ starting at $(0, 0)$ and a negative orbit $\{N_n\}$ starting at $(-1, 2)$.

Both sequences satisfy linear recurrences driven by multiplication by the Gaussian integer $1+i$. We show that the paired points remain symmetric with respect to the fixed midpoint $M = (-\frac{1}{2}, 1)$ for all n, satisfying the invariant $P_n + N_n = (-1, 2)$. Extending the iteration backward under the associated inverse maps produces the attractor of the well-known Twindragon fractal. In addition, the paired points yield the normalized quadratic sequence

$$a(n) = \frac{|P_n|^2 + |N_n|^2}{5} = 2^{n+1} + 1 - 2\,\mathrm{Re}((1 + i)^n),$$

which is always an integer and appears as A396151 in the OEIS. The construction therefore provides a new lattice-based viewpoint on the geometry and dynamics of the Twindragon.

1 Introduction

While experimenting with paired unit squares on graph paper, the author observed that repeatedly constructing new squares outwardly on the hypotenuse of the isosceles right triangle formed at each step produced two interlocking spiral sequences of lattice points. At every iteration, the two unit squares are positioned as point reflections of each other through their shared corner. These sequences remain on $\mathbb{Z}^2$ indefinitely, satisfy clean affine recurrences, and are related by a simple point-reflection symmetry through the non-lattice point $M = (-\frac{1}{2}, 1)$.

When the iteration is extended backward ($n < 0$), both orbits spiral inward toward M. The inverse maps turn out to be closely related to the two contractions that generate the famous Twindragon fractal (also known as the Twindragon or Davis dragon). Thus, a purely geometric, ruler and compass style construction on the integer lattice recovers one of the classic self-similar fractals in the plane.

This note presents the construction in full detail, derives the affine recurrences and their closed forms in the complex plane, proves the main invariants (including the new integer sequence $a(n)$), and clarifies the connection to the Twindragon attractor.

2 Geometric Construction

2.1 The Lattice

The construction is built entirely on the integer lattice $\mathbb{Z}^2$. Any point in the lattice can be chosen as the initial corner $P_0 = (0, 0)$; this choice is a labeling convention enabled by the translational symmetry of the lattice.

2.2 Base Iteration

At the base iteration $n = 0$, the construction consists of two unit squares that share exactly one common corner, denoted $P_0 = (0, 0)$. These two squares are positioned as point reflections of each other through the shared corner. This configuration establishes the initial paired-square structure from which all subsequent iterations are generated.

2.3 Corner Sum Identity

At each iteration $n \geq 0$, a pair of unit squares shares a common corner, denoted P_n (positive orbit) or N_n (negative orbit). The iteration point P_n (or N_n) is the average of the eight corner coordinates of the paired squares:

$$P_n = \frac{1}{8}\sum_{i=1}^{8}(x_i, y_i).$$

Because the two unit squares are point reflections of each other through the shared corner P_n, their eight corner coordinates are symmetric with respect to P_n. Therefore, P_n is exactly the centroid (average) of those eight points. This corner sum identity holds for every iteration.

2.4 Hypotenuse Line Construction

The two outer edges adjacent to the shared corner form the legs of an isosceles right triangle with the right angle at the shared corner. The hypotenuse construction at step n is the line containing the hypotenuse of this triangle. A new square is constructed outwardly on this hypotenuse line, and its far corner becomes the next iteration point P_{n+1} (positive orbit) or N_{n+1} (negative orbit).

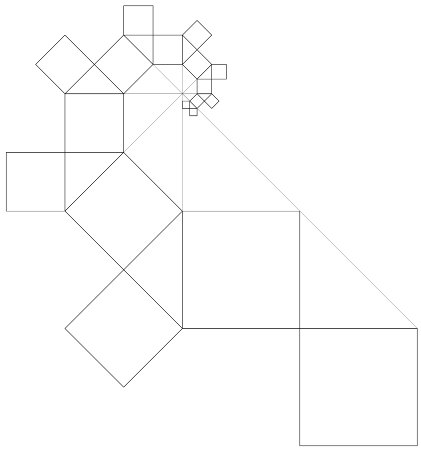

Figure 1: Positive orbit with hypotenuse line construction.

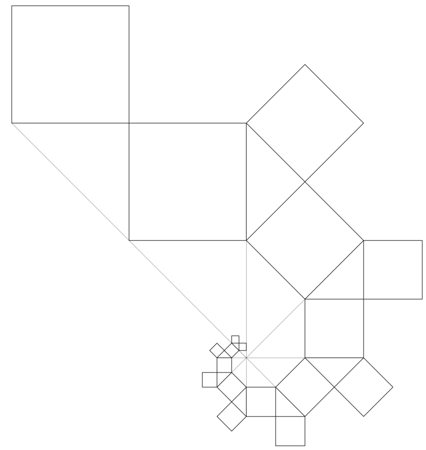

Figure 2: Negative orbit with hypotenuse line construction.

The hypotenuse lines generated by the positive orbit are concurrent at the point $N_0 = (-1, 2)$. This common intersection point naturally defines the starting point of the negative orbit. All hypotenuse lines of the positive orbit pass through N_0, while all hypotenuse lines of the negative orbit pass through $P_0 = (0, 0)$. In addition, all such lines fall into four families of parallel lines (corresponding to the four distinct directions produced by repeated $45°$ rotations) with perpendicular separations $\frac{1}{\sqrt{2}}$, 1, 2, and $\frac{3}{\sqrt{2}}$.

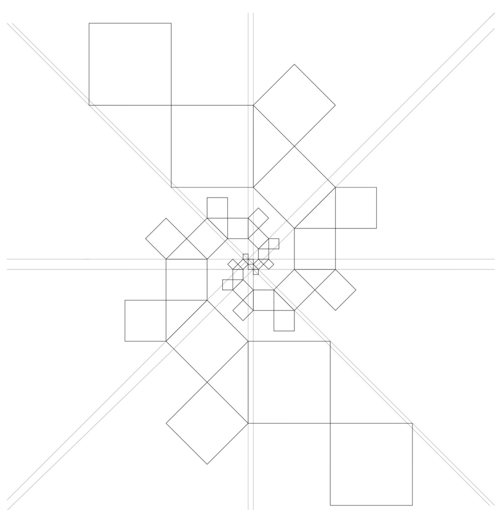

Figure 3: Positive and negative orbits on $\mathbb{Z}^2$ and the four families of parallel hypotenuse lines.

To connect the geometric construction to the algebra, consider the base case at $P_0 =$

$(0,0)$. The paired unit squares have outer edges from $(0,0)$ to $(1,0)$ and from $(0,0)$ to $(0,1)$ that form an isosceles right triangle whose hypotenuse runs from $(0,1)$ to $(1,0)$, with direction vector $(1,-1)$. Rotating this vector 90° counterclockwise produces the outward perpendicular $(1,1)$. Adding it to the hypotenuse endpoints yields the new corners $(1,2)$ and $(2,1)$. The construction selects $P_1 = (2,1)$ as the next iteration point of the positive orbit.

This single step is precisely the affine transformation that will be written below using the matrix A and translation vector v. The same local rule generates the full positive orbit on $\mathbb{Z}^2$.

Each step scales side lengths by $\sqrt{2}$ and rotates vectors counterclockwise by $\pi/4$. The construction therefore has 8-step rotational periodicity: after eight iterations the orientation repeats while all displacements are scaled by 16 about N_0.

This iterative process generates two sequences of iteration points on $\mathbb{Z}^2$: the positive orbit $\{P_n\}$ starting at $(0,0)$ and the negative orbit $\{N_n\}$ starting at $(-1,2)$.

2.5 Affine Recurrences

The first nine iteration points of the positive orbit are: $(0,0)_0$, $(2,1)_1$, $(3,4)_2$, $(1,8)_3$, $(-5,10)_4$, $(-13,6)_5$, $(-17,-6)_6$, $(-9,-22)_7$, $(15,-30)_8$, ...

The first nine iteration points of the negative orbit are: $(-1,2)_0$, $(-3,1)_1$, $(-4,-2)_2$, $(-2,-6)_3$, $(4,-8)_4$, $(12,-4)_5$, $(16,8)_6$, $(8,24)_7$, $(-16,32)_8$, ...

Both sequences satisfy linear recurrences on $\mathbb{Z}^2$. Define the matrix

$$A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

and the vector

$$v = (2,1).$$

The positive orbit satisfies the affine recurrence

$$P_{n+1} = AP_n + v, \quad \text{with } P_0 = (0,0).$$

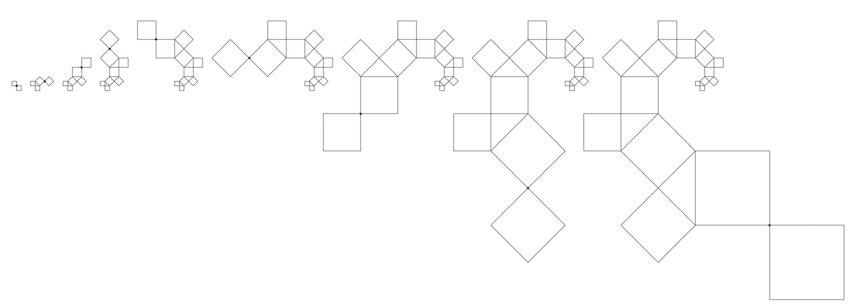

Figure 4: Iterations $n = 0$ to 8 of the positive orbit using paired squares.

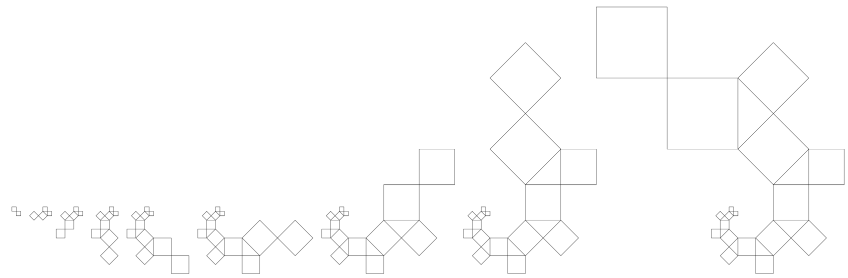

Figure 5: Iterations $n = 0$ to 8 of the negative orbit using paired squares.

The negative orbit satisfies the linear recurrence

$$N_{n+1} = AN_n, \quad \text{with } N_0 = (-1, 2).$$

Identifying $\mathbb{R}^2$ with the complex plane $\mathbb{C}$ via the map $(x, y) \mapsto x + iy$, the matrix A corresponds to multiplication by the Gaussian integer $1 + i$. Under this identification the recurrences become

$$P_{n+1} = (1 + i)P_n + (2 + i), \quad \text{with } P_0 = 0,$$

and

$$N_n = (-1 + 2i) \cdot (1 + i)^n, \quad \text{with } N_0 = -1 + 2i.$$

Solving these recurrences yields the closed forms

$$P_n = (-1 + 2i)\big[1 - (1 + i)^n\big], \quad N_n = (-1 + 2i) \cdot (1 + i)^n.$$

Multiplication by $1 + i$ rotates vectors counterclockwise by $\pi/4$ and scales their lengths by $\sqrt{2}$.

3 Fixed Midpoint M

The positive and negative orbits are symmetric with respect to a fixed point in the plane. For all $n \geq 0$ the iteration points satisfy the invariant relation

$$P_n + N_n = (-1, 2).$$

Dividing by 2 gives the midpoint

$$M = \big(-\frac{1}{2}, 1\big).$$

This point is independent of n. Consequently, N_n is the point reflection of P_n through M.

M is the only point in the construction that does not lie on the integer lattice $\mathbb{Z}^2$. All iteration points P_n and N_n, as well as all corners of the squares, lie on $\mathbb{Z}^2$.

The points $P_0 = (0, 0)$, $M = (-\frac{1}{2}, 1)$, and $N_0 = (-1, 2)$ are collinear along the direction $(-1, 2)$, and the distances satisfy $|P_0M| = |MN_0| = \sqrt{5}/2$.

Extending the recurrence backward ($n < 0$) causes both orbits to spiral inward toward M. As $n \to -\infty$ the points approach M. This backward behaviour is the contracting dynamics whose attractor is the Twindragon fractal.

4 Connection to the Twindragon Fractal

The forward dynamics of the construction are governed by the affine maps

$$T(z) = (1 + i)z + (2 + i), \qquad S(z) = (1 + i)z,$$

with $P_0 = 0$ and $N_0 = -1 + 2i$.

The corresponding inverse maps are

$$T^{-1}(z) = \frac{z - (2 + i)}{1 + i}, \qquad S^{-1}(z) = \frac{z}{1 + i}.$$

These inverse maps are affinely related to the standard contractions that generate the Twindragon fractal. In particular, $S^{-1}(z)$ corresponds to one of the canonical contractions under standard normalization, while $T^{-1}(z)$ differs from the second by a fixed translation. As a result, the iterated function system generated by $\{T^{-1}, S^{-1}\}$ is affinely equivalent (after a suitable change of coordinates) to the well-known IFS

$$f_1(z) = \frac{z}{1 + i}, \quad f_2(z) = 1 - \frac{z}{1 + i}$$

whose attractor is the Twindragon fractal (Davis and Knuth, 1970; Barnsley, 1988).

When the recurrence is extended to negative indices ($n < 0$), the points P_n and N_n are obtained by finite compositions of the inverse maps T^{-1} and S^{-1}. These backward iterations therefore generate points within the Twindragon attractor.

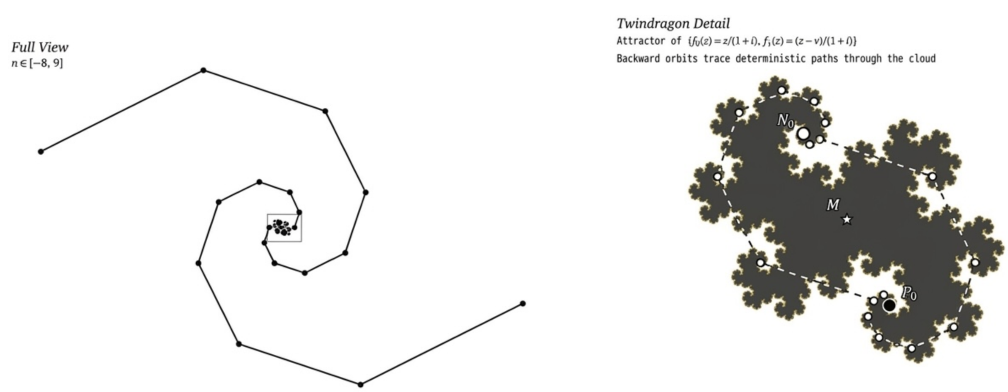

Figure 6: Bidirectional view: forward orbits ($n \geq 0$) expand outward on the integer lattice $\mathbb{Z}^2$, while backward orbits ($n < 0$) generate points within the Twindragon attractor.

5 Main Invariants

Theorem 5.1 (Sum Invariant)**.** *For all $n \geq 0$,*

$$P_n + N_n = (-1, 2).$$

Proof. Substituting the closed forms gives

$$P_n + N_n = (-1 + 2i)[1 - (1 + i)^n] + (-1 + 2i)(1 + i)^n.$$

The terms involving $(1 + i)^n$ cancel, leaving $-1 + 2i$. □

Theorem 5.2 (Norm Sum Invariant)**.** *The paired points P_n and N_n at each iteration generate the normalized quadratic quantity*

$$a(n) = \frac{|P_n|^2 + |N_n|^2}{5},$$

where $|U|^2 = x^2 + y^2$ denotes the squared Euclidean norm. This quantity is always an integer.

Proof. It satisfies the closed form

$$a(n) = 2^{n+1} + 1 - 2\,\mathrm{Re}((1 + i)^n),$$

where Re denotes the real part. It also satisfies the recurrence

$$a(n) = 2a(n - 1) - 2a(n - 2) + 2^n + 1$$

with initial conditions $a(0) = 1$, $a(1) = 3$.

To see that $a(n)$ is always an integer, observe that $(1+i)^n$ is a Gaussian integer for every n (its real and imaginary parts are integers by the binomial theorem). Therefore $\mathrm{Re}((1+i)^n)$ is an integer, and the closed-form expression is visibly an integer.

This sequence appears as A396151 in the OEIS. □

6 Conclusion

This note introduces a simple geometric construction on the integer lattice $\mathbb{Z}^2$ using paired unit squares. The process generates two interlocking affine spiral orbits that remain on lattice points, satisfy clean recurrences driven by the Gaussian integer $1 + i$, and are symmetric through the fixed midpoint $M = (-\frac{1}{2}, 1)$. It also produces the new integer sequence $a(n)$ (OEIS A396151). Extending the iteration backward reveals that the same maps generate the Twindragon fractal as their attractor. The construction therefore offers a concrete, visual lattice viewpoint on the dynamics of this well-known fractal and suggests natural directions for further exploration.

References

[1] M. F. Barnsley, *Fractals Everywhere*, Academic Press, 1988.

[2] C. Davis and D. E. Knuth, Number representations and dragon curves, *J. Recreational Math.* **3** (1970), 66–81 and 133–149.

[3] W. J. Gilbert, Fractal geometry derived from complex bases, *Math. Intelligencer* **4** (1982), 78–86.

[4] N. J. A. Sloane (ed.), The On-Line Encyclopedia of Integer Sequences, Sequence A396151, `https://oeis.org/A396151`.

Appendix A: Coordinate Tables for Paired Squares

To verify the corner-sum identity, we list all eight corner coordinates of the paired unit squares for each iteration from $n = 0$ to $n = 8$.

Because the two squares are always mirror images of each other across their shared corner P_n (or N_n), the average of those eight corners is exactly the shared corner point itself.

The tables below show the corner coordinates, their sums, and the final average (total sum divided by 8) for both the positive and negative orbits. All calculations were done by adding the coordinates directly.

Positive Orbit Paired Squares

Iteration point	**Pair of squares (corners)**	**Coordinate sums**	**Average (÷8)**
$P_0 = (0,0)$	a: $(0,0),(-1,0),(0,1),(-1,1)$	a: $(-2,2)$	$(0,0)$
	b: $(0,0),(1,0),(0,-1),(1,-1)$	b: $(2,-2)$	
$P_1 = (2,1)$	c: $(2,1),(1,0),(0,1),(1,2)$	c: $(4,4)$	$(2,1)$
	d: $(2,1),(3,0),(4,1),(3,2)$	d: $(12,4)$	
$P_2 = (3,4)$	e: $(3,4),(3,2),(1,2),(1,4)$	e: $(8,12)$	$(3,4)$
	f: $(3,4),(5,4),(3,6),(5,6)$	f: $(16,20)$	
$P_3 = (1,8)$	g: $(1,8),(1,4),(-1,6),(3,6)$	g: $(4,24)$	$(1,8)$
	h: $(1,8),(-1,10),(3,10),(1,12)$	h: $(4,40)$	
$P_4 = (-5,10)$	i: $(-5,10),(-1,6),(-1,10),(-5,6)$	i: $(-12,32)$	$(-5,10)$
	j: $(-5,10),(-9,10),(-9,14),(-5,14)$	j: $(-28,48)$	
$P_5 = (-13,6)$	k: $(-13,6),(-9,2),(-5,6),(-9,10)$	k: $(-36,24)$	$(-13,6)$
	l: $(-13,6),(-17,2),(-21,6),(-17,10)$	l: $(-68,24)$	
$P_6 = (-17,-6)$	m: $(-17,-6),(-17,2),(-9,2),(-9,-6)$	m: $(-52,-8)$	$(-17,-6)$
	n: $(-17,-6),(-17,-14),(-25,-14),(-25,-6)$	n: $(-84,-40)$	
$P_7 = (-9,-22)$	o: $(-9,-22),(-9,-6),(-1,-14),(-17,-14)$	o: $(-36,-56)$	$(-9,-22)$
	p: $(-9,-22),(-1,-30),(-9,-38),(-17,-30)$	p: $(-36,-120)$	
$P_8 = (15,-30)$	q: $(15,-30),(15,-14),(-1,-14),(-1,-30)$	q: $(28,-88)$	$(15,-30)$
	r: $(15,-30),(31,-30),(31,-46),(15,-46)$	r: $(92,-152)$	

Negative Orbit Paired Squares

Iteration point	**Pair of squares (corners)**	**Coordinate sums**	**Average (÷8)**
$N_0 = (-1, 2)$	-a: $(-1, 2), (0, 2), (-1, 1), (0, 1)$	-a: $(-2, 6)$	$(-1, 2)$
	-b: $(-1, 2), (-2, 2), (-1, 3), (-2, 3)$	-b: $(-6, 10)$	
$N_1 = (-3, 1)$	-c: $(-3, 1), (-2, 2), (-1, 1), (-2, 0)$	-c: $(-8, 4)$	$(-3, 1)$
	-d: $(-3, 1), (-4, 2), (-5, 1), (-4, 0)$	-d: $(-16, 4)$	
$N_2 = (-4, -2)$	-e: $(-4, -2), (-4, 0), (-2, 0), (-2, -2)$	-e: $(-12, -4)$	$(-4, -2)$
	-f: $(-4, -2), (-6, -2), (-4, -4), (-6, -4)$	-f: $(-20, -12)$	
$N_3 = (-2, -6)$	-g: $(-2, -6), (-2, -2), (0, -4), (-4, -4)$	-g: $(-8, -16)$	$(-2, -6)$
	-h: $(-2, -6), (0, -8), (-4, -8), (-2, -10)$	-h: $(-8, -32)$	
$N_4 = (4, -8)$	-i: $(4, -8), (0, -4), (0, -8), (4, -4)$	-i: $(8, -24)$	$(4, -8)$
	-j: $(4, -8), (8, -8), (8, -12), (4, -12)$	-j: $(24, -40)$	
$N_5 = (12, -4)$	-k: $(12, -4), (6, 0), (4, -4), (6, -8)$	-k: $(32, -16)$	$(12, -4)$
	-l: $(12, -4), (16, 0), (20, -4), (16, -8)$	-l: $(64, -16)$	
$N_6 = (16, 8)$	-m: $(16, 8), (16, 0), (8, 0), (8, 8)$	-m: $(48, 16)$	$(16, 8)$
	-n: $(16, 8), (16, 16), (24, 16), (24, 8)$	-n: $(80, 48)$	
$N_7 = (8, 24)$	-o: $(8, 24), (8, 8), (0, 16), (16, 16)$	-o: $(32, 64)$	$(8, 24)$
	-p: $(8, 24), (0, 32), (8, 40), (16, 32)$	-p: $(32, 128)$	
$N_8 = (-16, 32)$	-q: $(-16, 32), (-16, 16), (0, 16), (0, 32)$	-q: $(-32, 96)$	$(-16, 32)$
	-r: $(-16, 32), (-32, 32), (-32, 48), (-16, 48)$	-r: $(-96, 160)$	

These explicit checks confirm the identity through $n = 8$. The same averaging process continues for all higher iterations by the inductive structure of the construction (each new paired-square configuration is again centrally symmetric about its shared corner). The identity can also be verified computationally for arbitrary n using the closed-form expressions or the affine recurrence.